

Test Report: Agitator Prototypes V1 & V2

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Project: Substrate Mixer and Portioning Device

Component: Agitator (Paddle Mixer)

Prototype Version: Model Scale 1:5 (V1: T-paddle vs. V2: V-Paddle)

1 PART I — TEST PLAN

1.1 Research Hypotheses

- **Mixing Efficiency:** A simple paddle geometry achieves spatial homogeneity and stable mass ratios within 6 Crankshaft Rotations (CSR).
- **Longitudinal Transport:** The agitator geometry is capable of transporting and homogenizing localized “point-loaded” substrate volumes across the full length of the container within 10 CSR.

1.2 Test Objectives

To evaluate and compare the mixing efficiency, lateral transport capabilities, and operational stability of the T-paddle (V1) and V-paddle (V2) designs.

1.3 Measurement Methodology

- **Setup:** 1:5 scale model, 50% fill level.
- **Substrate:** coffee grounds (fine) and chives (large).

Table 1: Target Substrate Composition and Mass Ratios

Experiment	Coffee (m_c) [g]	Chives (m_{ch}) [g]	Target Ratio (R_m)
Test 1 (Uniform)	228	38	≈ 6.0
Test 2 (Localized)	230	57	≈ 4.0

- **Test 1 (Uniform Loading):** Components pre-distributed; sampled at 3 and 6 CSR.
- **Test 2 (Center Loading):** Chives placed exclusively in the center; sampled at 10 CSR to evaluate axial transport.



Figure 1: Experimental test setup for the 1:5 scale agitator model.

2 PART II — TEST EXECUTION DATA

2.1 Test 1: Uniform Initial Distribution

Table 2: V1 (T-paddle) Mass Distribution and Statistical Homogeneity

CSR	Position	Chives [g]	Coffee [g]	Ratio (R_m)	St. Dev. (s)
3	Left	< 1.0	9.0	9.00	3.53
3	Middle	2.0	8.5	4.25	
3	Right	3.0	6.0	2.00	
6	Left	3.0	9.0	3.00	0.87
6	Middle	2.0	9.0	4.50	
6	Right	3.0	9.0	3.00	

Target $R_{target,1} = 6.0$. Standard deviation s indicates homogeneity (lower is better).

Table 3: V2 (V-paddle) Mass Distribution and Statistical Homogeneity

CSR	Position	Chives [g]	Coffee [g]	Ratio (R_m)	St. Dev. (s)
3	Left	2.0	9.5	4.75	1.04
3	Middle	3.0	8.0	2.67	
3	Right	1.5	5.5	3.67	
6	Left	2.5	6.5	2.60	0.94
6	Middle	2.0	8.0	4.00	
6	Right	2.5	11.0	4.40	

2.2 Test 2: Localized (Center) Initial Distribution

Table 4: Test 2: Axial Transport Homogeneity after 10 CSR

Prototype	Position	Ratio (R_m)	Avg. $\bar{R}_m$	St. Dev. (s)
V1 (T)	Left / Mid / Right	4.80 / 3.00 / 7.67	5.16	2.35
V2 (V)	Left / Mid / Right	5.25 / 3.40 / 6.50	5.05	1.56

Target $R_{target,2} = 4.0$. Lower s indicates better axial distribution.

3 PART III — TEST REVIEW

3.1 Observations

- **Operational Stability:** V1 exhibited smooth operation. V2 created a "shovel effect" that increased the substrate's kinetic energy, leading to material ejection (splattering) from the container.
- **Mixing Speed:** V2 reached visual homogeneity slightly faster (4 CSR) than V1 (6 CSR).
- **Lateral Transport:** Both prototypes showed limited axial mixing. In Test 2, concentrations remained significantly higher at the loading point (center) even after 10 CSR.

3.2 Interpretation & Measurement Uncertainty

A digital scale with a precision of ± 1 g was used. At a mass range of 1 g to 9 g, a 1 g deviation results in a relative error of up to 50%, explaining the fluctuations in R_m . Furthermore, increasing substrate moisture led to granulation, hindering clean separation during sampling. Visual observation remains the primary metric for this scale, which was positive for both designs. Industry research confirms that T-paddle geometries are a proven standard for full-scale agitators.

3.3 Hypothesis Evaluation

Partially Confirmed. Spatial uniformity was achieved during pre-distributed loading (Test 1). However, target mass ratios were not fully recovered due to segregation and measurement limits. Longitudinal transport (Test 2) was insufficient for full homogeneity.

3.4 Final Recommendations

- **Selection:** The **T-paddle (V1)** is selected for full-scale implementation due to superior consistency, manufacturing simplicity, and the absence of loss of substrate material while stirring.
- **Design:** Short "V-wings" may be added to the welded T-bars for structural reinforcement if needed.
- **User Guideline:** Users should distribute substrate components evenly across the container during loading to compensate for limited lateral transport.



Figure 2: Agitator V1 (T-paddle) in an unloaded container condition



Figure 3: Agitator V2 (V-paddle) filled with test substrate and after 10 crank shaft rotations